SAKILA MOVIE RENTAL ANALYSIS REPORT

MECE Framework-Based Business Intelligence Analysis

1. Executive Summary

The Sakila Movie Rental Analysis Dashboard was developed to analyze rental business performance, customer behavior, inventory trends, payment patterns, geographic revenue distribution, and store operations using Power BI.

A MECE (Mutually Exclusive, Collectively Exhaustive) framework was applied to ensure the analysis was:

- Structured
- Non-overlapping
- Comprehensive
- Business-oriented

The dashboard provides a centralized executive reporting solution that enables stakeholders to monitor operational efficiency, customer engagement, rental performance, inventory utilization, and regional revenue contribution.

The project combines data modeling, DAX calculations, dashboard storytelling, KPI reporting, and interactive business intelligence techniques.

2. Objective of the Project

The primary objective of this project is to:

- Analyze movie rental business performance
- Identify revenue-generating factors
- Understand customer behavior and rental trends
- Monitor inventory and film category performance
- Evaluate geographic and store-wise business contribution
- Build an executive-level interactive dashboard for decision-making

3. About the Dataset

The project uses the Sakila Movie Rental Database, which simulates a DVD rental business.

The dataset contains information related to:

- Customers
- Rentals
- Payments
- Movies
- Film Categories
- Inventory
- Staff
- Stores
- Geographic Locations

The data was imported into SQL Workbench and connected to Power BI for transformation, analysis, and dashboard development.

4. MECE Framework Analysis

The entire business analysis was divided into four mutually exclusive and collectively exhaustive sections.

A. CUSTOMER & SALES ANALYSIS

This section focuses on customer purchasing behavior, payment trends, and revenue contribution.

Key Metrics Analyzed

- Total Revenue
- Total Customers
- Monthly Revenue Trend
- Payment Method Distribution
- Customer Segment Contribution
- Customer Ratings by Location

Insights

1. Online payments contributed 33.69% of total rental revenue, making it the leading payment method.

The payment method analysis shows that online payments contributed the largest share of rental transactions compared to cash and credit card payments.

Business Impact

- Indicates increasing customer preference for digital transactions
- Demonstrates convenience-driven customer behavior
- Suggests the business should strengthen digital payment systems

2. Active customers contributed approximately 66K in sales, while inactive customers contributed less than 2K.

The customer segmentation analysis revealed that active customers are responsible for nearly all revenue generation.

Business Impact

- Active customer retention is critical for business sustainability
- Loyalty programs can increase repeat rentals
- Marketing campaigns should focus on customer retention

3. Monthly revenue showed strong growth during later months.

The revenue trend visual indicates significant growth in rental revenue over time.

Business Impact

- Indicates improving rental demand
- Suggests successful customer engagement
- Helps management identify high-performing periods

4. Customer ratings remain stable around average levels.

The customer rating analysis demonstrates relatively stable ratings across different regions.

Business Impact

- Reflects consistent customer experience
- Suggests stable service quality
- Provides opportunity for service enhancement to achieve higher ratings

B. FILM & INVENTORY ANALYSIS

This section analyzes movie inventory, film categories, rental duration, and content distribution.

Key Metrics Analyzed

- Film Categories
- Rental Duration Distribution
- Film Rating Distribution
- Language Distribution
- Average Rental Rate
- Average Rental Duration

Insights

1. Action films recorded at the highest average rental rate of 3.0 among all film categories.

The dashboard indicates that Action films generated the highest rental activity and rental rate performance.

Business Impact

- High demand for action-based content
- Inventory expansion in action genre may increase revenue
- Marketing efforts can prioritize top-performing genres

2. Long-duration rentals dominate movie rentals.

The rental duration distribution shows that long rental durations contributed the highest proportion of rentals.

Business Impact

- Customers prefer extended viewing flexibility
- Longer rental periods improve customer convenience
- Rental pricing strategies can be optimized accordingly

3. PG-13 films have the highest inventory count.

The inventory distribution by film rating shows PG-13 movies dominate the inventory.

Business Impact

- Indicates strong demand for family-friendly entertainment
- Balanced audience targeting strategy
- Helps inventory management prioritize popular ratings

4. Film categories maintain consistent rental rates.

Most categories display similar rental pricing structures.

Business Impact

- Reflects standardized pricing strategy
- Prevents pricing inconsistencies
- Simplifies rental pricing management

C. STAFF & STORE PERFORMANCE ANALYSIS

This section evaluates store operations, employee contribution, and staff-related business performance.

Key Metrics Analyzed

- Store Revenue by Country
- Staff Employment Distribution
- Average Rental Duration by Staff
- Customer Ratings by Location

Insights

1. Store performance varies significantly across countries.

Certain countries contribute considerably more revenue than others.

Business Impact

- High-performing locations should receive greater investment
- Underperforming locations may require operational improvements
- Helps optimize regional expansion strategy

2. Staff rental duration performance remains consistent.

The average rental duration by staff member appears balanced.

Business Impact

- Indicates operational consistency
- Suggests uniform employee performance
- Helps maintain service quality standards

3. Employment distribution between staff categories is balanced.

The employment duration distribution indicates a relatively balanced workforce structure.

Business Impact

- Ensures operational stability
- Prevents workforce dependency risks
- Improves staff workload management

D. GEOGRAPHIC REVENUE ANALYSIS

This section focuses on revenue generation across countries and global rental performance.

Key Metrics Analyzed

- Revenue by Country
- Top Revenue Generating Countries
- Bottom Revenue Generating Countries
- Geographic Revenue Map

Insights

1. India generated the highest rental revenue at approximately 2.1K, followed by the United States at 1.8K.

The geographic revenue analysis shows India generated the highest rental revenue.

Business Impact

- India represents the strongest market opportunity
- Regional promotions can further increase revenue
- Geographic expansion strategies can prioritize similar markets

2. Several countries contribute minimal revenue.

Bottom-performing countries generated significantly lower revenue compared to top-performing regions.

Business Impact

- Highlights areas requiring marketing improvements
- Helps identify low-demand regions
- Assists in operational cost optimization

3. Revenue distribution is globally diversified.

The map visualization demonstrates worldwide customer engagement.

Business Impact

- Reduces dependency on a single market
- Strengthens business scalability
- Supports international business opportunities

5. Dashboard Features

The Power BI dashboard includes:

- Interactive slicers
- Page navigation buttons
- KPI cards
- Dynamic filtering
- Geographic mapping
- Revenue trend analysis
- Insight storytelling section
- Executive reporting layout

6. DAX Measures Used

Several DAX measures were created to support dashboard calculations.

Examples

Total Revenue

```
Total Revenue = SUM(payment[amount])
```

Total Customers

```
Total Customers = DISTINCTCOUNT(customer[customer_id])
```

Average Customer Rating

```
Avg Customer Rating = AVERAGE(rating[rating])
```

7. Business Recommendations

Based on the analysis, the following recommendations are suggested:

Customer Strategy

- Increase loyalty and retention programs
- Strengthen online payment infrastructure
- Improve customer engagement campaigns

Inventory Strategy

- Expand high-performing film categories like Action
- Maintain balanced inventory for popular film ratings
- Monitor rental duration trends for pricing optimization

Geographic Strategy

- Invest more in high-revenue countries
- Develop targeted campaigns for low-performing regions
- Use geographic insights for expansion planning

Operational Strategy

- Maintain balanced staff distribution
- Improve store performance monitoring
- Use KPI tracking for operational efficiency

8. Conclusion

The Sakila Movie Rental Dashboard successfully transformed raw rental data into meaningful business intelligence insights using Power BI.

The MECE framework ensured the analysis remained:

- Structured
- Complete
- Non-overlapping
- Easy to interpret

The dashboard provides decision-makers with a centralized platform to monitor business performance, customer trends, operational efficiency, and geographic revenue contribution.

This project demonstrates practical skills in:

- SQL
- Data Cleaning
- Data Modeling
- DAX
- Power BI Dashboard Design
- Business Intelligence Reporting
- Analytical Storytelling

9. Tools & Technologies Used

- SQL Workbench
- Excel
- Power BI
- DAX
- Power Query
- Sakila Database

10. Outcome

The final solution is a professional executive-level reporting dashboard capable of:

- Tracking revenue performance
- Monitoring customer behavior
- Evaluating inventory utilization
- Analyzing geographic business contribution
- Supporting strategic business decisions

The project highlights the ability to convert transactional data into actionable business insights using modern business intelligence tools.